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Reproducing Adiyatullin et al. 2017 on single-sweep HBT data —project report

Date: 2026-08-10 ·**Status:** reproduction campaign complete (null results, quantified) ·**Audience:** collaborators / group meeting ·**PDF:** 26-08-10-project-report.pdf ·**中文版:** 26-08-10-project-report-zh.md

This report closes the attempt to reproduce the two core measurements of Adiyatullin et al., "Periodic squeezing in a polariton Josephson junction", *Nat. Commun.* **8**:1329 (2017), on our own streak-camera data. It is organized as a funnel: first what the data is (§2), how it is processed (§3), and how the processing is checked and where its limits lie (§4–§5); then the reproduction result (§6), the dynamics the data *does* contain (§7), and conclusions (§8). Validation records and method comparisons are collected in the appendices.

Bottom line up front

- **Instrument.** We built a single-sweep, photon-resolved Hanbury Brown–Twiss (HBT) analysis chain for streak-camera frames of a polariton Josephson dimer: per-frame time-zero from a laser reference pulse (constant-fraction discrimination, CFD), per-photon event extraction, spatial band pairing of the two condensates × two HBT arms, and the paper's own normally ordered $g^{(2)}$ estimator. Every stage is checked on controlled inputs (§4).
- **Reproduction target.** The paper's Fig. 2(a) —anti-phase population alternation between the two condensates (the paper's mesas) at a ∼10 ps period —and Fig. 2(c) —zero-delay $g^{(2)}$ switching between Poisson (1) and bunched (>1) in step with that oscillation (§1).
- **Result.** Neither observable reproduces on our data —across two independent acquisition batches and two independent event curations. For the population observable the null is decisive: the upper limits sit far below the paper's full-contrast signal. For $g^{(2)}$ the window-pooled statistic is Poisson-consistent in both arms, and any reference-locked oscillation is bounded below the top of the paper's amplitude range. All numbers: §6; one-figure summary: §8.

- **Checks behind the nulls.** The same chain recovers injected paper-type signals end-to-end; the decision criteria were registered before the data was analyzed; and the longer independent batch tightens every upper limit without changing any conclusion (§4, §6.3).
- **What the data does show.** The Josephson dynamics is present as *relative-phase winding*: tilted two-condensate interference fringes in the k-space synchroscan images, oscillating with a clean period in every recorded frame. In this detuned/running-phase regime the populations barely modulate —consistent with, not contradictory to, the real-space null (§7).
- **What it would take.** Covering the *lower* half of the paper′ s $g^{(2)}$ amplitude range calls for a substantially longer acquisition —or, more effectively, a lower detector background (§5 quantifies the trade). The curated event tables and count cubes for both batches are archived and ready for the follow-up 2-D $g^{(2)}(t_1,t_2)$ and cross-condensate analyses (§8).

1. Introduction and the reproduction target

1.1 The physics in one paragraph each

Polariton Josephson dimer. Exciton-polaritons are light–matter quasiparticles in a semiconductor microcavity; confined in two coupled circular mesas (2 μm diameter, 2.5 μm center spacing, tunnel coupling J = 0.4 meV in the paper′ s sample), two condensates exchange population the way two coupled superfluids do: the population imbalance z = (N_L − N_R)/(N_L + N_R) oscillates at a period set by the coupling (∼10 ps in the paper), damped by the polariton lifetime (5.2 ps). This is the **Josephson oscillation** the paper images directly.

HBT and $g^{(2)}$. A Hanbury Brown–Twiss setup splits the emission on a 50:50 beamsplitter and detects the two arms independently; the zero-delay second-order correlation $g^{(2)}(0) = \langle n_1 n_2\rangle/(\langle n_1\rangle\langle n_2\rangle)$ distinguishes Poissonian emission ($g^{(2)} = 1$) from bunched emission ($g^{(2)} > 1$). Because the estimator is normally ordered and the two arms dilute the same photon stream independently, arm-efficiency asymmetry cancels exactly in the ratio —no arm calibration is needed. The paper′ s Fig. 2(c) reports $g^{(2)}(0)(t)$ of each mesa switching between Poisson and bunched *in phase with the Josephson oscillation*, the two mesas in anti-phase, with the bunching peaks growing from ≈1.2–1.5 to $\gtrsim$2 as the population decays.

What the paper claims beyond that —and what we do not touch. The paper′ s Fig. 2(d)(e) squeezing curves are derived from a fitted model, not measured; intensity bunching by itself is not a quadrature-squeezing measurement (that would need a phase-locked local oscillator and balanced homodyne detection, which neither setup has). This campaign reproduces the *intensity* and *intensity-correlation* measurements only; we make no squeezing claim in either direction.

1.2 The target figures

Paper Fig. 2(a): accumulated intensity of the left (blue) and right (red) mesa, 0–30 ps, log scale. Anti-phase alternation at ~10 ps period under a decaying envelope; the gray band marks the SNR<20 region. Average over >3.9×10⁶ laser pulses.

Paper Fig. 2(c): $g^{(2)}(0)(t)$ of the two mesas (shaded ±1 standard error of the mean, SEM) switching between Poisson and bunched in step with the population oscillation, peaks growing with time.

1.3 What "reproduction" means here

The paper′ s measurement facts (from its Supplementary Note 1): >3.9×10⁶ pulses over 34 h; custom photon counting on a synchroscan streak camera acting as the single-photon detector for both HBT arms; time bins Δt = 1.296 ps = 2 detector px (0.648 ps/px); per-pulse time jitter corrected with a split-off laser

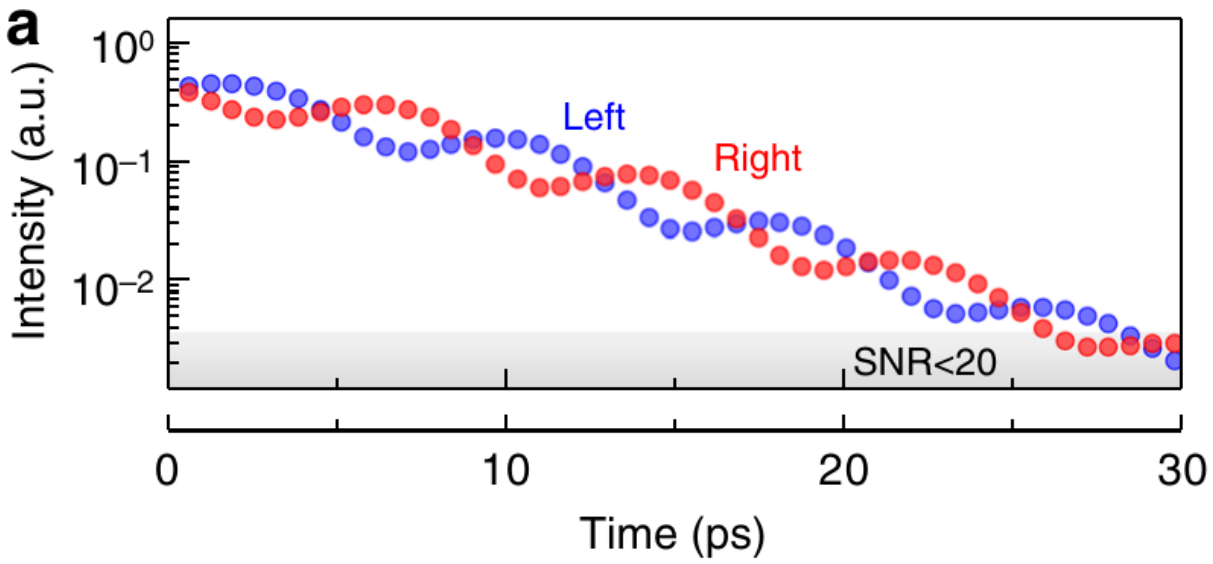


Figure 1: Adiyatullin 2017 Fig. 2(a) original

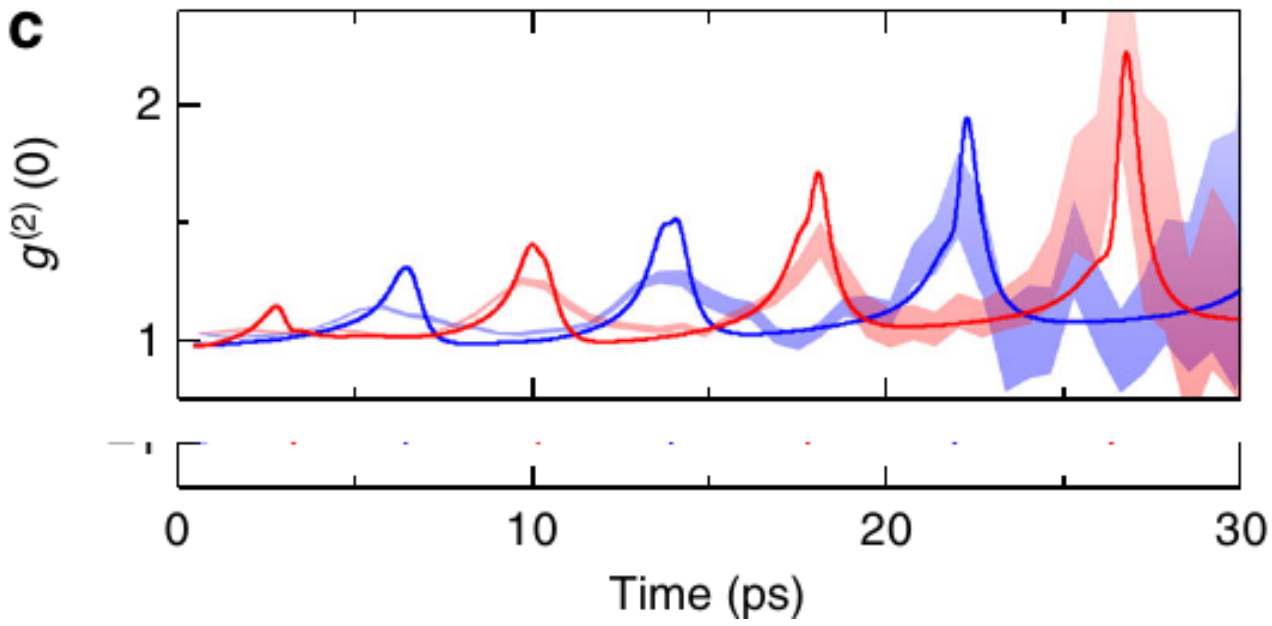


Figure 2: Adiyatullin 2017 Fig. 2(c) original

reference pulse (algorithm not published); intensity per mesa I = n_L + n_R summed over arms; $g^{(2)}$ from per-pulse counts with SEM-propagated errors; 3.4 ps FWHM instrument resolution.

Our excitation regime is different: **nonresonant reservoir-fed condensates triggered by an optical-Stark pulse**, imaged in real space over a 1 ns window, versus the paper′ s resonant bonding/antibonding pulse and 30 ps window. We therefore replicate the *measurement pipeline and the observable form*, on the closest observables our platform produces, and we state regime differences as boundaries (§8) rather than papering over them. Where the paper left method choices undocumented (time-zero algorithm, photon threshold, background treatment, L/R mapping), we made our own and checked them —the choices and their tests are listed in §3 and Appendix B.

Datasets covered. The campaign runs on the 26-08-04 HBT acquisition family (main batch + a 3.6× longer supplement batch of the same condition); the earlier 26-07-31 single-sweep dataset (2 × 1000 frames) served as the precursor where the estimator and the statistics-first policy were developed (§6.4).

2. Data: what the raw material looks like

2.1 Datasets

Dataset	Content	Frames	Role
`26-07-31-single-sweep`	streak frames, 2 conditions (`ZeroOrderPowerRatio` 0.05 / 0.18)	2 × 1000	precursor (§6.4)
`26-08-04-hbt-time-reference`, main condition `hbt_condensate_josephson_regime`	single-sweep real-space HBT frames with laser time reference on the side	20 000 (19 992 kept; 8 frames fail time-zero)	primary
same acquisition, 8 further conditions	synchroscan k-space / real-space images, no-pulse intensity calibrations, backgrounds, focus cross-sections, delay calibration	10 each	calibration (the k-space condition drives §7)
supplement batch of the same HBT condition ("New folder (10)")	identical acquisition metadata except sweep length; signal ≈65 % of main batch	72 000 (71 971 kept; 29 fail)	independent confirmation (§6.3)

All frames are 1016 × 1344 px uint16 streak-camera images: the **vertical axis is time** (1 ns over 1016 px ⇒ 0.984 ps/px, row 0 = window start, per the acquisition metadata conventions in `metadata-conventions.toml`); the **horizontal axis is uncalibrated detector pixels**. Raw MAT files are immutable; analysis reads lossless derived HDF5 copies with per-frame provenance and SHA-256, pixel-exact verified against the sources (supplement: 498 stratified frames, 0 mismatches).

2.2 Anatomy of the measurement

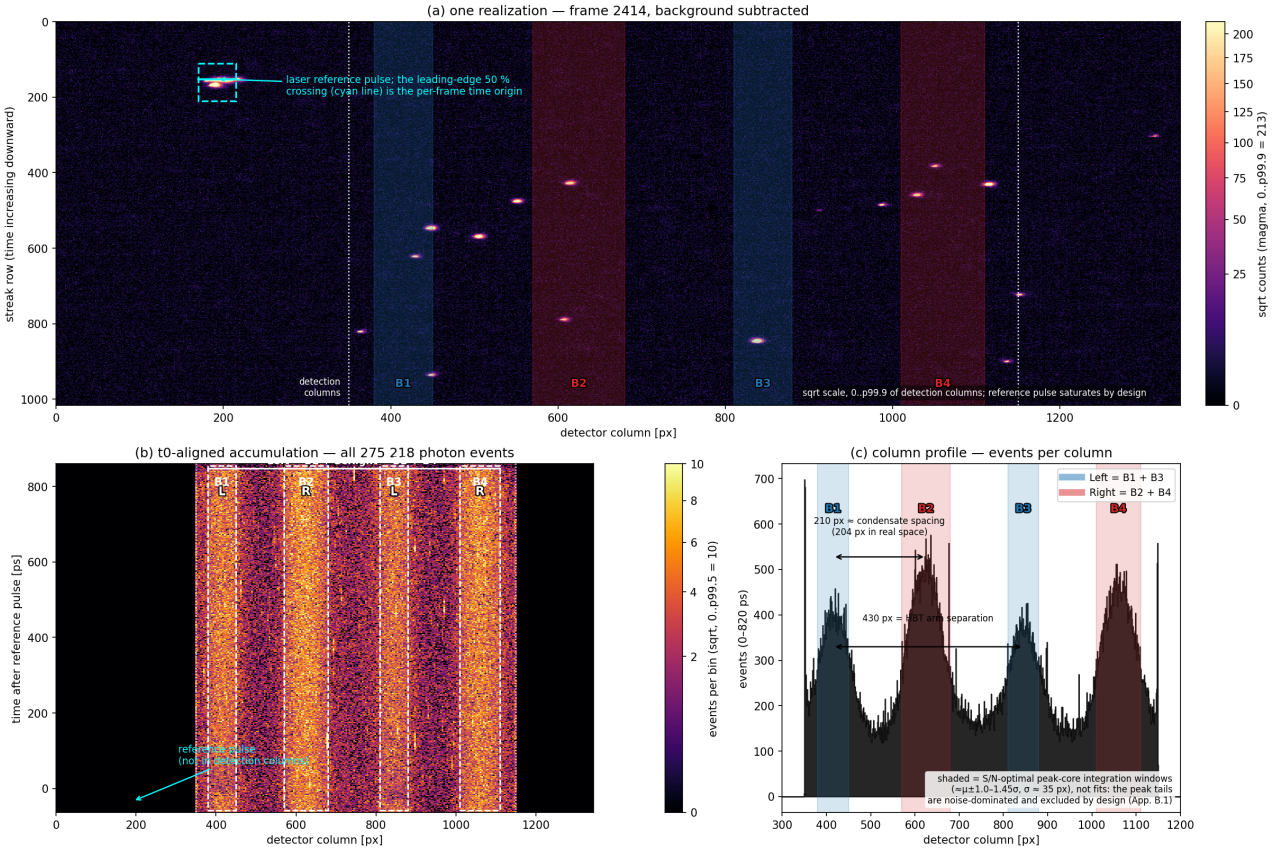


Figure 3: Data preview

Figure 3 (C1). (a) One realization: a single background-subtracted frame (sqrt intensity scale). Emission arrives as sparse single-photon spots on a ~105 counts/px pedestal; the bright laser reference pulse at the left edge provides each frame' s time origin —the cyan line marks the leading-edge 50 % crossing (CFD50, §3 step 3) that defines it for this frame. (b) All 275 218 detected photon events of the main batch after per-frame alignment: four vertical emission bands emerge. (c) Column profile of the aligned events: the four bands B1–B4 and their pairing into Left = B1+B3 and Right = B2+B4 (each condensate is imaged by both HBT arms; the 210 px B1↔B2 spacing matches the 204 px real-space condensate spacing from the no-pulse calibration, and the 430 px B1↔B3 spacing is the arm separation). The shaded windows are peak-core integration windows, deliberately narrower than the full peak widths —the tails add mostly residual noise (S/N optimization, Appendix B.1).

Three structural facts matter for everything downstream:

1. **Four bands, two condensates, two arms.** On the single-sweep frames each condensate appears once per HBT arm, giving four emission bands; pairing them as L = {B1, B3}, R = {B2, B4} reproduces the no-pulse real-space intensity ratio (0.586/0.579 per arm vs 0.578 calibration), while the cross pairing gives inconsistent ratios (0.59/1.51) and is rejected.
2. **A per-frame time reference.** The split-off laser pulse lands on the same frame (cols ~170–215, outside the detection columns). The sweep' s two-state wander is ~13 ps rms —far larger than the analysis bin width —so per-frame alignment to this reference is not optional. The sample-pulse/reference-pulse delay was measured at 0.32 ± 0.20 ps, i.e. consistent with zero offset.

3. **A measured background.** A dedicated background condition (unexposed streak sweep) gives a per-pixel pedestal of ≈105 counts, subtracted frame-by-frame; its applicability to the supplement batch was verified directly (median off-signal difference +0.02 counts/px). What remains after subtraction is Poisson-like residual detector noise that also triggers the photon detector —quantified in §5.

3. From frames to physics: the measurement pipeline

The chain turns 20 000 raw frames into accumulated intensity traces and a per-photon event table, and from there into $g^{(2)}(0)(t)$. Each stage below states what is done and the physical reason; how the stage is checked is the subject of §4, and the full validation records are in Appendix A.

1. **Conversion.** MATLAB v7.3 exports → per-experiment HDF5 (schema 2), keeping only the streak-camera payload, with per-source-file provenance and pixel-exact verification. *Reason:* immutable raws stay untouched; every derived number traces to a hashed source frame.
2. **Background subtraction.** The measured per-pixel background (≈105 counts) is subtracted from every frame. *Reason:* the streak pedestal is large compared to single-photon spots; using the *measured* background (not a fitted one) avoids injecting model bias into the photon detector.
3. **Per-frame time zero (CFD50).** On the reference pulse' s leading edge, the 50 %-of- peak crossing row, sub-row interpolated —a constant-fraction discriminator (CFD), the standard answer to amplitude walk and pulse-shape breathing. *Reason:* the reference pulse' s shape breathes frame to frame; a centroid or fixed-threshold crossing tracks that breathing (+2.7 ps bias, 1.3 ps/frame scatter), the CFD does not (§4, Appendix B.2). 8 of 20 000 frames (0.04 %) fail the signal-quality cut and are excluded and recorded.
4. **Photon-event detection.** 8-connected bright components above 25 counts (integrated, post-background) and ≥2 px in detection columns 350–1150, each reduced to a count-weighted sub-pixel centroid. The threshold sits at the knee of a sensitivity scan. *Reason:* one photon triggers a multi-pixel microchannel-plate (MCP) cloud; component-wise integration recovers one event per photon.
5. **Band windows and pairing.** The four emission bands are located on the photon-event column histogram (out-of-band floor is Poisson, so a 5σ sub-window threshold with gap bridging finds the bands; the analog profile' s lumpy fixed-pattern baseline cannot be thresholded this way). Windows: B1 380–450, B2 570–680, B3 810–880, B4 1010–1110; pairing L = {B1,B3}, R = {B2,B4} per the geometry checks of §2.2. The chosen windows sit within 0.6 % of the signal-to-noise optimum (Appendix B.1).
6. **Products.** Per band: accumulated mean/SEM traces after sub-row alignment to the per-frame time zero, in two parallel sources —*photon* (event counts) and *analog* (integrated counts). Per dataset: a curated per-photon **event table** (frame, sub-pixel x/y, jitter-corrected $\tau = y - t0$, amplitude; 275 218 events main, 661 090 supplement) and (frames × 4 bands × time-bin) **count cubes** at 1/2/4 px binning. The event table and the traces come from a *single* detection pass, with an automatic table↔traces reconstruction check at the end of every extraction.
7. **The $g^{(2)}$ estimator.** For each condensate, cross-arm coincidences of its two bands (L: B1×B3; R: B2×B4) per time bin. Each pairing correlates a condensate *with itself* —the two arms are two independent copies of the same emission after the beamsplitter —so L and R are the auto-correlations of condensate 1 and condensate 2 respectively; no condensate-1×condensate-2 cross-correlation enters this report (archived as follow-up, §8.3). $g^{(2)}(0)(t)$ comes from per-frame counts n_1, n_2 with the paper' s normally ordered ratio, plus a residual-background correction β estimated from out-of-band events per time bin (β removes the noise contribution to the mean; the variance cost is irreducible, §5). Displayed per-bin error bands use a null-analytic ("plug-in") SEM —at $\sim 10^{-3}$

occupancy per bin the Gaussian error bar of a single bin is meaningless, so the band shows the experiment's resolution around the Poisson line; window-pooled values use leave-one-frame-out jackknife SEMs (Monte-Carlo-calibrated, Appendix A.3).

8. **Statistical policy.** Two rules were fixed before the real-data $g^{(2)}$ was ever computed: a **power gate** (compute what the dataset can resolve, from measured rates, before running the estimator; if the paper's range is not covered, the deliverable degrades to an upper-limit report —it did), and **pre-registered decision criteria** (four tests with surrogate null distributions, written down 2026-08-06 ahead of the computation). Any positive result would additionally have had to survive injection recovery (§4) —none was triggered.

4. Recovery of known signals

The check on the chain above is whether it recovers signals we planted —end-to-end, from synthetic frames through the untouched production code.

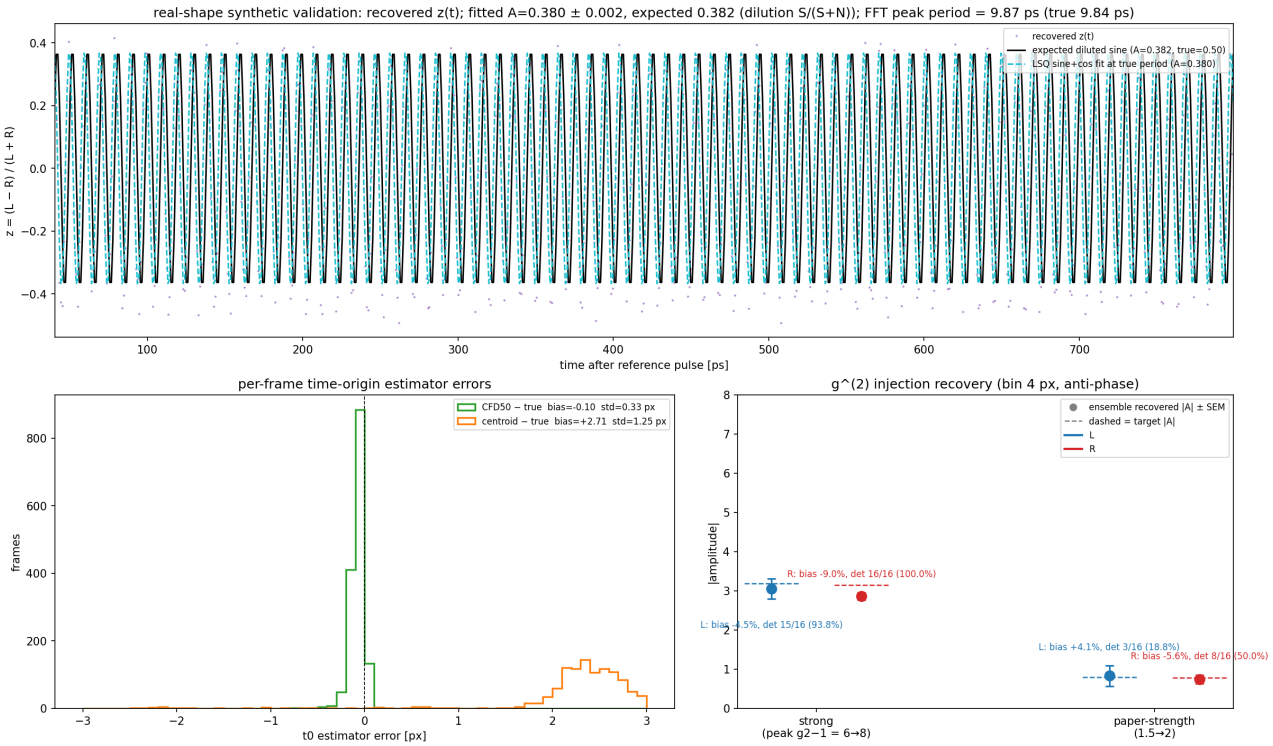


Figure 4: Injection recovery

*Figure 4 (C2). Left: end-to-end recovery of an injected anti-phase population oscillation (period 9.84 ps —the paper's Josephson period —z amplitude 0.5, locked to the true per-frame time zero) in synthetic frames that carry **measured** reference-pulse shapes with their real frame-to-frame breathing, the measured background level, and the observed two-state time-base wander. Top: the recovered z(t) oscillates at 9.87 ps with the expected (noise-diluted) amplitude. Bottom: per-frame time-zero error —CFD50 is unbiased (−0.10 ± 0.33 px); the former centroid estimator is biased +2.71 px with 1.25 px scatter because it tracks the pulse-shape breathing, which is why the pipeline uses CFD50. Right: cube-level $g^{(2)}$ injection —a paper-style modulation (9.84 ps, anti-phase, growing envelope) added to the real measured rates and residual background: the ensemble amplitude bias stays within the pre-set 10 % acceptance criterion at both injection strengths; the single-replica detection rate at paper strength (L 19 %, R 50 %) is the direct empirical reading of the power gate's "borderline" verdict (§5).*

Three complements, detailed in the appendices:

- **Frame-level injection through the curated "dataset A" chain** (real event patches planted into real frames): 98.8–99.6 % of injected blobs survive curation with zero time-zero perturbation, and the injected 9.84 ps modulation is recovered at the expected amplitude (Appendix A.4).
- **Estimator bias checks on synthetic inputs**: on synthetic Poisson cubes the $g^{(2)}$ estimator returns 1 within Monte-Carlo tolerance, and the analytic SEMs match 64-replica Monte Carlo within 4 % at the working bin width (Appendix A.3).
- **Robustness to method choices**: the conclusions are identical under the alternative time-zero estimator, under the alternative (four-Gaussian) band-window semantics, and under the alternative (global-mean) background model —each comparison is quantified in Appendix B.

5. Error analysis and resolving power

Knowing precisely what the measurement *cannot* see is what makes the nulls of §6 informative. The error budget decomposes as follows (numbers for the main batch; the supplement scales accordingly).

Event localization (from the 26-07-31 precursor study, same detector): 0.16–0.17 px RMS per axis, dominated by a deterministic sub-pixel window-truncation bias (0.13 px), not by detector noise (0.002 px); artificial-spot injection completeness 92 %. Irrelevant under the 1–4 px analysis binning.

Time-zero. Per-frame sweep wander is 13.2 ps rms (two-state, real synchroscan behavior —the two timing estimators agree on it to 0.1 ps). After CFD50 alignment the residual estimator noise is 0.33 px ≈ 0.32 ps, far below the 4 px (3.94 ps) $g^{(2)}$ bin.

Statistical budget for $g^{(2)}$. In the coincidence-counting limit,

$$\mathrm{SEM}[g^{(2)}(0)] \approx 1/\sqrt{(F \cdot s_1 \cdot s_2)} \times 1/\sqrt{(\mathrm{purity}_1 \cdot \mathrm{purity}_2)},$$

with F frames, per-arm signal rate s, and purity = signal/(signal+noise). Measured:

factor	value	consequence
frames F	19 992 (main) / 71 971 (supplement)	SEM ∝ $1/\sqrt{F}$
signal rate s	0.50–1.21 events/frame/arm (0–820 ps)	quadratic lever
residual detector noise	**45–55 % of in-band events**	SEM ×2.0–2.3 ⇔ losing 4–5× frames
bin width × period	sinc penalty 1.32× at 9.84 ps, ≈1 at 74.4 ps (4 px bins)	harmless at our (longer) periods
time-zero residual	0.32 ps	negligible

The noise fraction is the second-largest lever and has **no analysis-side remedy** —signal and noise events are the same MCP-cloud objects (amplitude/size distributions coincide; the best per-event discriminator achieves 1.7:1 against a 3:1 break-even), raising the detection threshold 25→70 counts removes 11 % noise at 6 % signal cost, and narrowing the windows gains 1–4 %. Only acquisition-side background reduction helps —and it helps quadratically, noise halving ≈ frames doubling.

Resolving power vs the paper. The figure below is the measurement's frequency- dependent 95 % sensitivity floor, computed from measured rates before the estimator ever ran (the power gate), Monte-Carlo-verified:

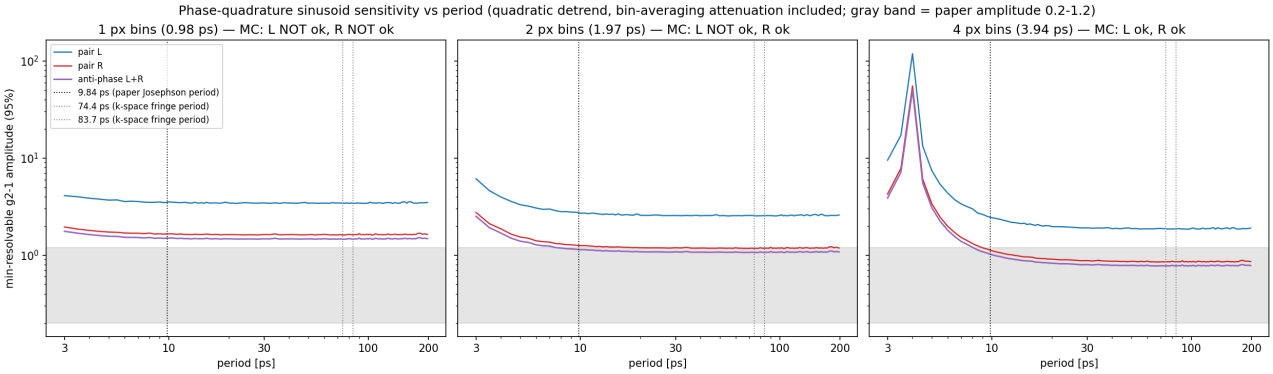


Figure 5: Sensitivity vs period

Figure 5 (C3). Minimum resolvable reference-locked $g^{(2)}$ oscillation amplitude (95 %) vs period for 1/2/4 px bins; gray band = the paper's Fig. 2(c) amplitude range ($g^{(2)}-1 \approx 0.2–1.2$). The 4 px width passes Monte-Carlo validation for both arms and is the working width. The dataset covers the upper half of the paper's range.

Frame-count outlook at current rates (4 px, anti-phase-combined median / at 9.84 ps): 20 k frames → 0.78 / 1.01 (measured: 0.81 / 1.09 —the gate was accurate); 72 k → 0.41 / 0.53; 92 k pooled → 0.36 / 0.47; **≈3×10⁵ frames → 0.20 / 0.26**, covering the paper's lower bound.

6. The reproduction result

With the pipeline's limits quantified, here is what it measures on the real data.

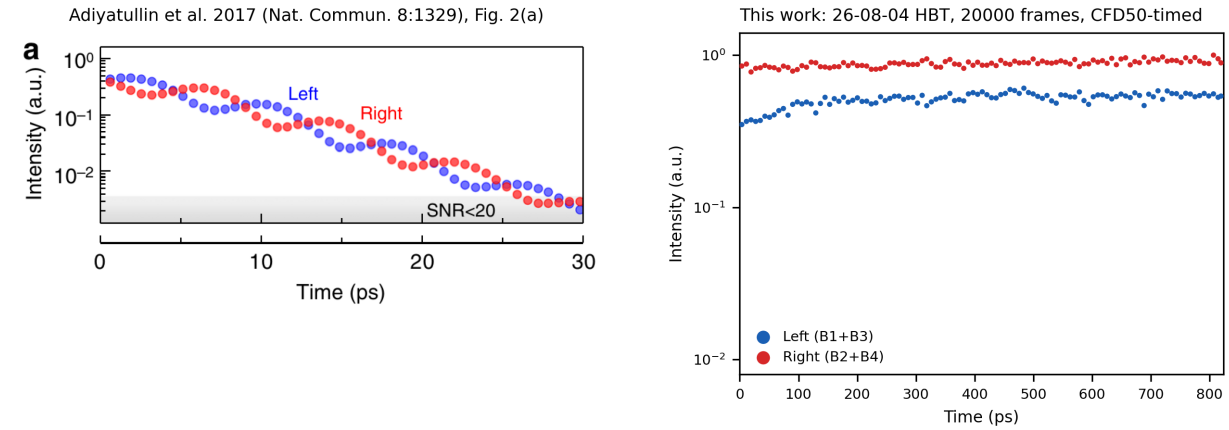
6.1 Population dynamics: no anti-phase alternation

Figure 6 (C4). The paper's Fig. 2(a) (left) vs this work (right): accumulated Left (B1+B3, blue) and Right (B2+B4, red) intensity, log scale, peak-normalized, 19 992 frames. The traces are flat; the early ($t \lesssim 150$ ps) left-side dip is a monotonic overdamped recovery, not an oscillation. The paper's full-contrast anti-phase alternation is absent. (The panel's baked-in caption quotes the scan as 0.008–0.019, median 0.012 —the definitive values are those in the text below.)

Quantitatively, on z = (I_L − I_R)/(I_L + I_R):

- Cross-arm detrended-fluctuation correlations are insignificant (r = 0.030, permutation p = 0.525; r = 0.095, p = 0.145 at two binnings).
- A reference-locked sinusoid scan over 3–200 ps periods gives a 95 % upper limit on the z amplitude of **0.0123 median / 0.0193 worst** —i.e. ≈1–2 % intensity modulation, against the paper's full-contrast measured signal (z ≈ ±0.4–0.5). Targeted fits at the a-priori periods: 9.84 ps |A| = 0.0041 (UL95 0.0116); 74.4 ps 0.0038 (0.0113); 83.7 ps 0.0090 (0.0165).
- The most sensitive direct look —subtracting each arm's mean and magnifying the plateau fluctuations with ±1 SEM bands (offset-zoom figure) —finds the bands overlapping everywhere and a weak **positive** common-mode correlation (r = +0.174, p = 0.023): slow shared drift, the opposite sign from anti-phase.

The scan behind these numbers: fitted z-oscillation amplitude and its $|A|+2\sigma$ 95 % upper limit vs period (left, log scale), and the detrended combined z(t) with its ±1 SEM band (right). No period stands out; the upper limit stays at the 1–2 % level everywhere.



Left: original Fig. 2(a) of Adiyatullin et al. 2017 (Nat. Commun. 8:1329) — accumulated emission intensity of the left (blue) and right (red) mesa on a logarithmic scale, showing anti-phase population oscillations; the gray band marks the low-confidence SNR<20 region. Right: this work — replication on the 26-08-04 HBT batch (20000 single-sweep frames; photon-event source, rebin 8 ≈ 7.9 ps/bin, peak-normalized). Each condensate is recorded simultaneously by both HBT arms; the plotted intensity sums its two arm bands (Left = B1+B3, Right = B2+B4). The per-frame time zero is calibrated by CFD50, leading-edge constant-fraction discrimination on the reference pulse (50%-of-peak crossing, sub-row interpolated), placing the origin ~2.7 ps earlier than the former half-max centroid definition; the measured StreakCamBackG background is subtracted. Outcome: no anti-phase oscillation — cross-arm z fluctuations are uncorrelated (r=0.03–0.10, p>0.14); the 95% upper limit on a phase-locked sinusoidal z amplitude is 0.008–0.019 over 3–200 ps periods (median 0.012; ~1–2% intensity modulation); the t≤150 ps left-side dip is an overdamped monotonic recovery, not an oscillation.

Figure 6: Fig. 2(a) replica

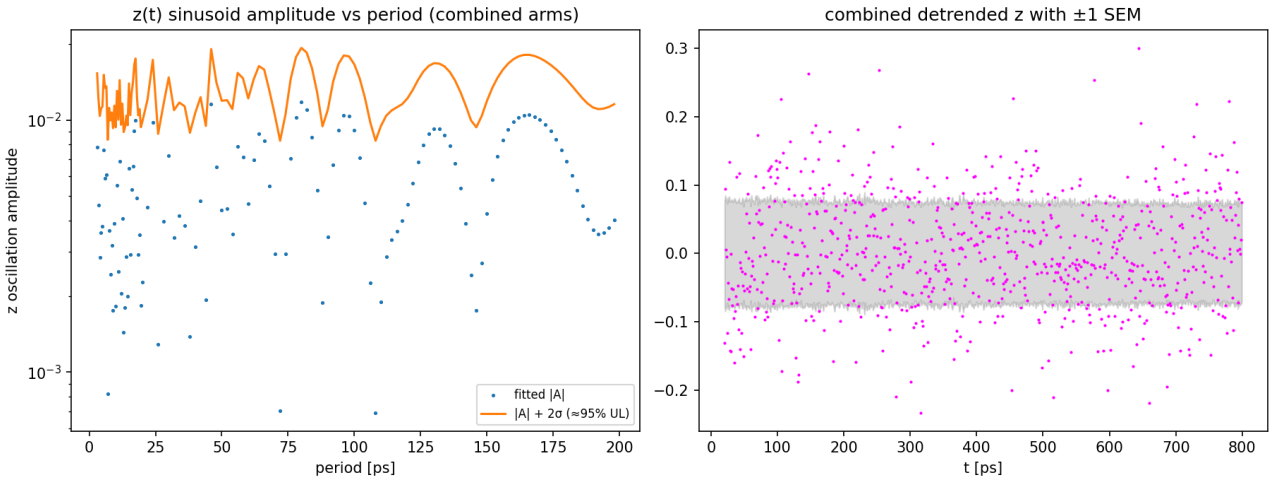
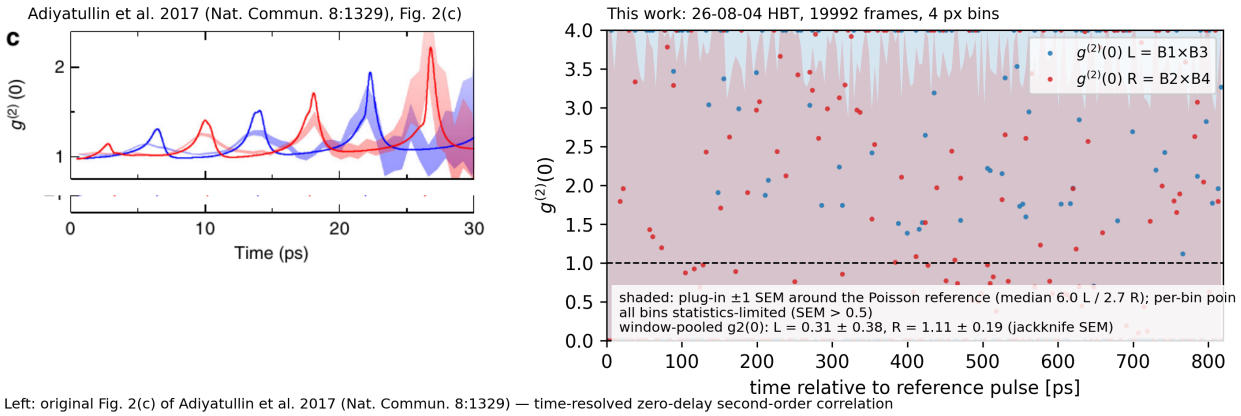


Figure 7: z upper-limit spectrum

6.2 Photon statistics: Poisson-consistent, no bunching switching



Left: original Fig. 2(c) of Adiyatullin et al. 2017 (Nat. Commun. 8:1329) — time-resolved zero-delay second-order correlation g(2)(0)(t) of the left (blue) and right (red) mesa; shaded areas are ±1 s.e.m.; the light statistics switches between Poissonian and bunched in phase with the Josephson oscillations, the two mesas in counterphase. Right: this work — same estimator (Suppl. Note 1) on the 26-08-04 HBT batch (19992 single-sweep frames, CFD50 time-zero, measured-background subtraction, 4 px bins). The per-bin regime is statistics-limited: ~50% of detected events are residual detector-noise clusters and the per-bin occupancy is ~1e-3 events/frame, so per-bin g2 is a heavy-tailed statistic. Points are per-bin estimates (clipped to [0, 4]); shaded bands are the plug-in ±1 s.e.m. resolution around the Poisson reference (Monte-Carlo validated), not error bars on the points. Outcome: no significant deviation from Poisson — window-pooled g2(0) = 0.31 ± 0.38 (L), 1.11 ± 0.19 (R) (jackknife s.e.m.); oscillation scan (3-200 ps, reference-lo (p = 0.15); 95% upper limits on the g2-1 amplitude: median 0.81, 1.09 at 9.84ps, 0.68 at 74.4ps, 0.54 at 83.7ps. With 20k frames only the upper part of the paper's amplit range (0.2-1.2) is resolvable; ~3e5 frames would be needed for 95% sensitivity to |A| = 0.2 (G3 power gate, docs/26-08-04-fig2c.md).

Figure 8: Fig. 2(c) replica

Figure 8 (C5). The paper's Fig. 2(c) (left) vs this work (right): $g^{(2)}(0)(t)$ of each condensate (L = B1×B3 blue, R = B2×B4 red), 4 px bins; the shaded band is the experiment's resolution (plug-in SEM, §3 step 7) around the Poisson line. The curve is statistics-limited per bin by construction; the information is in the pooled value and the oscillation limits.

- **Window-pooled $g^{(2)}(0)$** (0–820 ps, 4 px, jackknife): **L = 0.310 ± 0.380**, **R = 1.114 ± 0.188** —z = −1.82 / +0.60, both consistent with Poisson (|z| < 2, pre-registered threshold).
- **The four pre-registered tests** (criteria fixed 2026-08-06, before the computation; 200 circular-shift surrogates): (a) no significantly bunched/antibunched bin (p = 0.32 / 0.86); (b) no significant reference-locked oscillation (combined p = 0.15) —95 % upper limit **0.812 median** over 3–200 ps, 1.086 at the paper's 9.84 ps period, 0.680 / 0.540 at the measured k-space periods 74.4 / 83.7 ps;
  (c) no anti-phase —the detrended L/R correlation is weakly *positive* (r = +0.100, p = 0.886 one-sided), mirroring §6.1's common-mode drift;
  (d) no envelope growth (late-minus-early pooled $g^{(2)}$−1: L +0.159, p = 0.45; R +0.176, p = 0.37).

6.3 An independent batch and an independent curation say the same thing

Supplement batch (71 971 frames, same condition, acquired back-to-back; analyzed stand-alone because its signal rate is ≈65 % of the main batch): the same flat traces (anti-phase UL95 median **0.0084**), pooled $g^{(2)}(0)$ L = 1.304 ± 0.320 / R = 0.990 ± 0.117 (Poisson-consistent), all four pre-registered tests null, oscillation UL95 median **0.496**. Every limit tightened; every conclusion held (fig2a / fig2c supplement panels).

Dataset A —a curated re-extraction keeping only events that can physically be condensate signal (in-band, off the one defective column, not touching frame edges; 161 357 of 275 218 events kept in the main batch, verified as an exact subset), with a per-band local noise floor replacing the global-mean β: identical conclusions with tighter limits (population UL95 0.0131, 0.0226 after noise-dilution correction; $g^{(2)}$ oscillation UL95 median 0.58; and for the supplement' s dataset A: population UL95 0.0083, $g^{(2)}$ oscillation UL95 median 0.488), and better paper-strength injection detection (L 31 % / R 81 %) from the purer noise model.

The nulls persist across the pipeline variant, the window choice, and the noise model —in everything tried here, they follow the data.

6.4 The precursor dataset (26-07-31)

The earlier single-sweep dataset (2 × 1000 frames, no time reference) is where the estimator was built: trackpy-based event localization (validated against 2-D Gaussian fits, 92 % within 0.2 px), four-Gaussian band attribution (μ±2σ acceptance), and the same Supplementary-Note-1 $g^{(2)}$ estimator. Its full $g^{(2)}$ matrix came out Poisson-consistent within ~1.5σ at 1000 frames —a statistics-limited null that closed that dataset and motivated two design decisions for the 26-08-04 campaign: acquire with a per-frame time reference, and run the power gate *before* the science computation.

7. The dynamics we do see: phase winding in k-space

The oscillation the collaborators expect is present in the apparatus —just not as real-space population modulation.

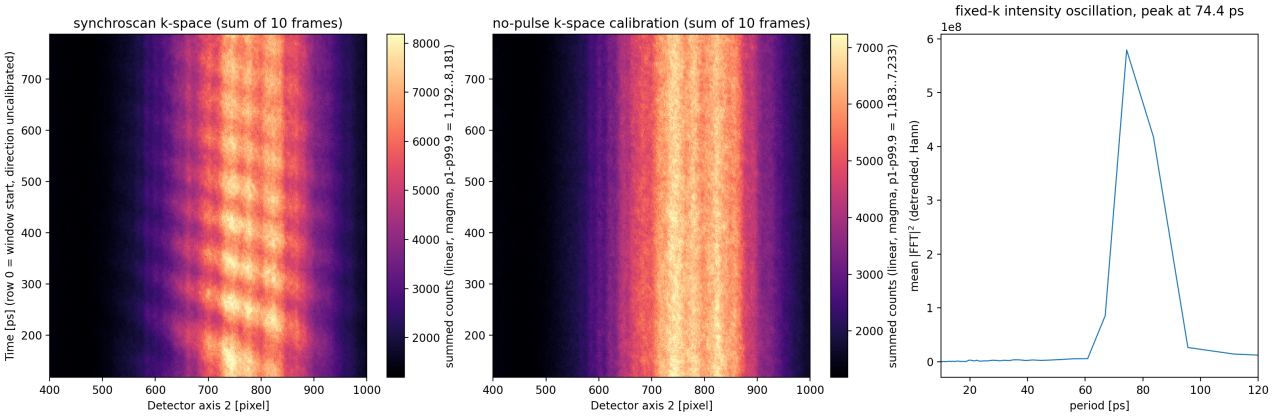


Figure 9: k-space fringes

Figure 9 (C6). Synchroscan k-space images (camera-internal accumulation, hence immune to the per-frame time-base wander): the two-condensate interference pattern forms a tilted checkerboard —fringes that tilt with time are a relative phase that winds with time. A fixed-k column oscillates with main period ***74.4 ps*** *(9 of 10 k-bins; secondary 83.7 ps), visible in each of the 10 frames individually (2-D FFT peak/median 11–18). The no-pulse k-space calibration shows only the smooth static emission envelope —no fringe pattern —excluding a detector origin of the fringes; 74.4 ps ≈ 13.4 GHz is far above known electronics pickup.*

Phase winding with barely-modulated populations is the signature of a detuned, running-phase Josephson regime —exactly the regime in which a Fig. 2(a)-type *population* observable stays flat. The k-space measurement and the real-space null are two views of the same state, and they agree. Resolving the

phase dynamics at the single-sweep level would need per-frame k-space imaging or per-frame likelihood analysis —a different acquisition, listed in §8.

8. Conclusions

Figure 10 (C7). All quantitative conclusions in one view. Top: population-oscillation 95 % upper limits (z-space) for the two batches and the two curations, against the paper's full-contrast band. Middle: $g^{(2)}$ oscillation 95 % upper limits against the paper's amplitude range. Bottom: window-pooled $g^{(2)}(0)$ with ±1 SEM against the Poisson line. Every upper limit sits far below the corresponding paper signal; every pooled value sits on Poisson within error.

8.1 Claim-by-claim account

Paper claim (observable)	This work	Verdict	Evidence
Fig. 2(a): anti-phase population alternation, ~10 ps, full contrast (measured z ≈ ±0.4–0.5)	flat traces; UL95 1.2 % (main) / 0.8 % (supplement) / 2.3 % (dataset A, dilution-corrected) of z	**Not reproduced — decisive** (paper signal ≳30× above limits)	§6.1, Fig. 6
Fig. 2(c): $g^{(2)}(0)(t)$ switches Poisson↔bunched in phase with the oscillation, peaks 1.2→≳2	pooled Poisson-consistent; oscillation UL95 0.81 / 0.50 / 0.58 / 0.49 by batch/curation; four pre-registered tests null	**Not reproduced within sensitivity** (limits cover the paper range's upper half)	§6.2, Fig. 8
Anti-phase between the two condensates (both observables)	weak *positive* common-mode correlation in both (r = +0.17 intensity; r = +0.10 $g^{(2)}$)	**Not reproduced; sign opposite**	§6.1–6.2
Bunching-envelope growth with time	late−early differences consistent with zero	**Not reproduced**	§6.2 (d)
Josephson dynamics exists in the device	confirmed —as 74.4 ps relative-phase winding in k-space, populations unmodulated	**Confirmed, different quadrature**	§7, Fig. 9

8.2 Boundaries (what this campaign does not say)

1. Different physical regime: resonant pulsed bonding/antibonding excitation, 30 ps window (paper) vs nonresonant reservoir-fed + Stark kick, 1 ns window (this work). We replicated the measurement pipeline and observable form, not the identical state.
2. $g^{(2)}(0) > 1$ would not have been quadrature squeezing anyway; the paper's Fig. 2(d)(e) squeezing curves are model-derived and were never in scope.
3. "In phase with Josephson oscillations" can only be referenced to the laser reference pulse here, since the population oscillation itself is absent in real space.

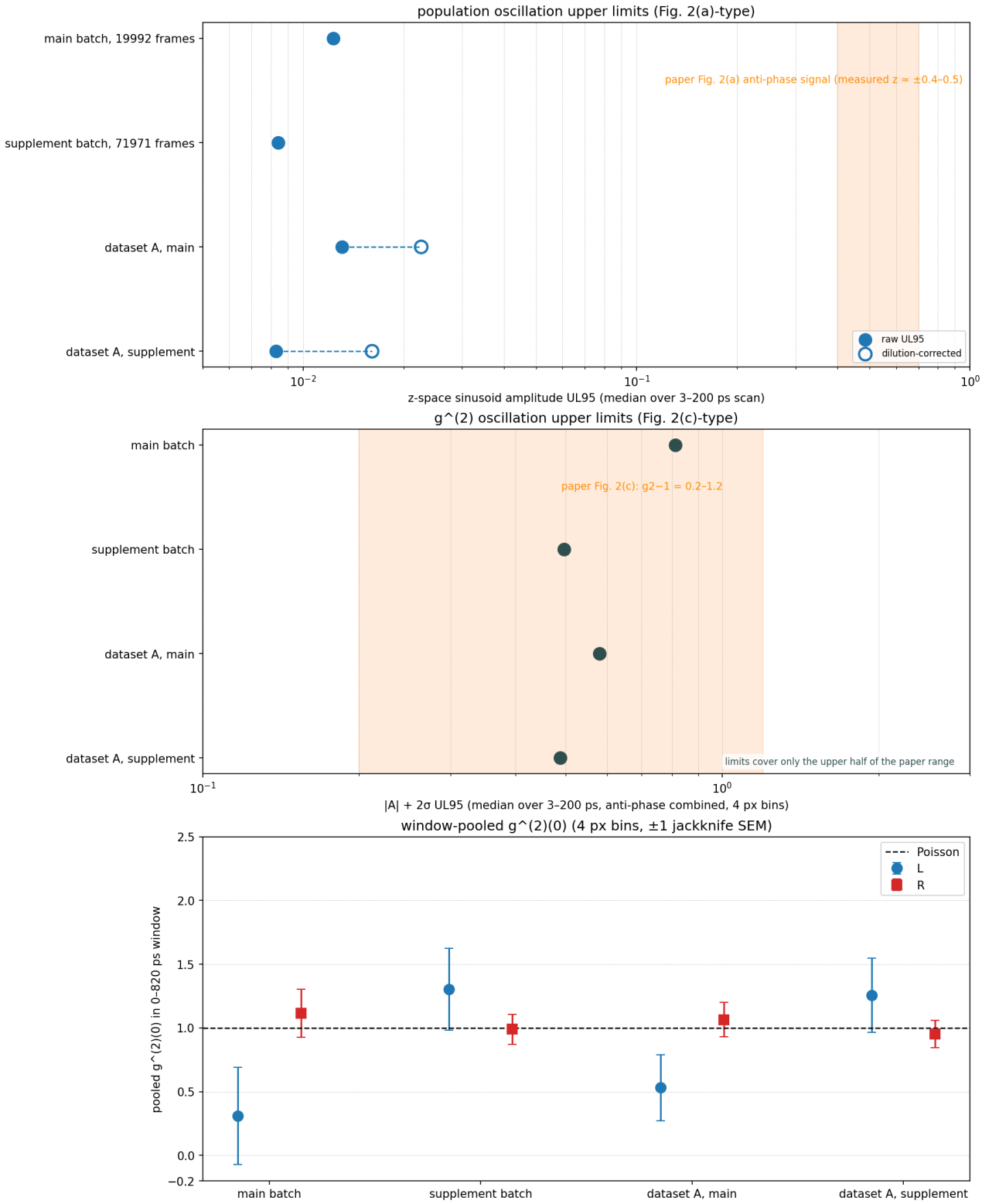


Figure 10: Results dashboard

4. The two batches differ physically (signal rate, L/R ratio) and were analyzed stand-alone, not pooled.
5. Nothing here bears on the 26-07-31 retry of Fig. 2(a) or on the single-kick trajectory question (per-frame/hierarchical analysis) —both remain open by design.

8.3 What comes next

- **2-D $g^{(2)}(t_1,t_2)$** (the paper's Fig. 3) and **cross-condensate** LR/RL correlations: the event tables and count cubes of both batches are already archived for exactly this.
- **Statistics**: ≈3×10^5 frames at current rates cover the paper's lower $g^{(2)}$ bound; halving the detector background is worth ≈2× frames (acquisition-side only).
- **Phase dynamics at single-sweep level**: per-frame k-space imaging or per-frame likelihood analysis, motivated by §7.
- **Open metrology gap**: the absolute row→ps calibration is metadata-based; an in-data calibration remains a to-do (it does not affect any conclusion here —all periods are measured in-frame).

Declaration. The reproduction campaign against Adiyatullin et al. 2017 Fig. 2(a) and Fig. 2(c) is **complete**: both observables were measured on two independent batches and two curations with a pipeline whose every stage was checked on controlled inputs (§4, Appendix A); the results are null with quantified upper limits; the instrument's resolving power is known and its shortfall against the paper's lower amplitude range is quantified with a concrete remedy.

Appendix A. Validation records

A.1 Pre-registration and the power gate

- The four science tests (bunching existence, oscillation, anti-phase, envelope growth), their statistics, surrogate construction, and significance thresholds were written into `DOCS/26-08-04-fig2c.md` §G6 on **2026-08-06, before any real-data $g^{(2)}$ computation**. Deviating from Poisson requires $|z| \geq 2$ (pooled) or surrogate-calibrated $p < 0.05$ (tests).
- The **power gate** (G3) ran before the full pipeline: from the measured band rates and the measured out-of-band residual-noise density (flat in τ, $\chi^2/\mathrm{ndf} = 1.16$) it predicted per-bin SEM medians, the pooled precision (L ±0.41 / R ±0.18 —measured jackknife: 0.38 / 0.19), and the oscillation sensitivity (median 0.78 —measured 0.81). Its "borderline" verdict (only the upper half of the paper's 0.2–1.2 range covered) triggered the pre-agreed downgrade to an upper-limit deliverable, confirmed with the user on 2026-08-06. The supplement batch's gate passed the same way (predicted 0.52, measured 0.50).

A.2 Surrogate null distributions

One arm's per-frame counts are circularly shifted by a random frame offset, breaking same-frame coincidences while preserving each arm's rate structure and the t0 structure; 200 surrogates pass through the identical statistic pipeline. Multiplicity across bins/periods is controlled by taking scan-maximum statistics ($\max|z|$, $\max|A|/\sigma$) on both data and surrogates.

The oscillation test at the working 4 px width: fitted |A| scans for both pairs and the anti-phase combination with the combined UL95 curve (left; dotted marks at the a-priori periods 9.84 / 74.4 / 83.7 ps), and the observed scan-maximum statistic against its surrogate null distribution (right; p = 0.154).

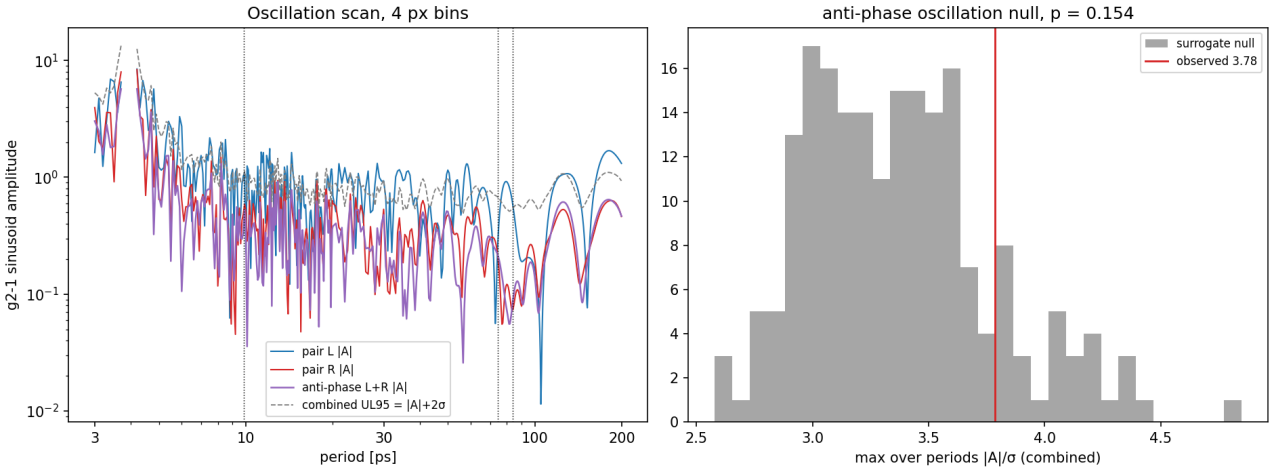


Figure 11: pre-registered test example

A.3 Monte-Carlo calibration of the analytic errors

64 synthetic Poisson replicas through the estimator chain: at the working 4 px width the analytic UL95 agrees with MC within 4 % for both pairs (ratios 1.04/1.01 —pass); 2 px passes for R (1.03) but not L (1.54 —heavy-tailed; used qualitatively only); 1 px fails (spike-dominated). Pooled jackknife SEMs sit within 0.86–1.07 of MC. Supplement: 4 px and 2 px ratios within [1.000, 1.024]; 1 px L fails identically. This calibration is why 4 px is the reporting width throughout.

A.4 Injection record (full)

test	injected	recovered	verdict
fig2a-style frame-level, real reference shapes (1500 synthetic frames)	z = 0.5 anti-phase, 9.84 ps	period 9.87 ps; t0 bias −0.10 px, std 0.33 px; amplitude at the noise-diluted expectation	pass (<2 % period error)
fig2c-style cube-level, strong (peak $g^{(2)}-1$ = 6→8), 16 replicas	9.84 ps anti-phase, growing envelope	ensemble bias L −4.5 %, R −9.0 %; period within 2 %: L 11/16, R 16/16; all replicas significant	pass (<10 % bias)
fig2c-style cube-level, paper strength (1.5→2), 16 replicas	same	ensemble unbiased (L +4.1 %, R −5.6 %); single-replica detection L 19 %, R 50 %	unbiased; partial detection = the borderline regime
supplement cube-level	same protocols	strong: L +0.2 %, R −4.1 %; paper: L −7.4 %, R +4.3 %, detection L 19 %, R 81 %	pass

test	injected	recovered	verdict
dataset A cube-level	same	strong: L −2.2 %, R −7.9 % (main), L −9.2 %, R −4.6 % (supplement); paper-strength detection L 31 %, R 81–100 %	pass; purity gain converts to detection rate
dataset A frame-level (real frames + 394 real event patches)	z-mode: 10 867 blobs in 2856 frames; bunching-mode: 252 blobs (126 cross-arm pairs)	retention 98.8 % / 99.6 %; zero t0 perturbation; recovered amplitudes match the dilution expectation within 10 %	pass

A.5 Extraction self-consistency

Event table and fig2a traces issue from a single detection pass; every full extraction ends with an automatic table→traces reconstruction check (integer-count criterion ≤2 counts/bin from float64 bin-edge ties: main batch [1,1,1,2] counts across the four bands —pass; supplement 14–39 counts/band = 0.015–0.044 % of band events, tolerance scaled accordingly —pass). Time-zero failures are excluded and recorded (main: frames 3543, 4458, 4589, 6614, 7935, 8190, 9765, 11225; supplement: 29 frames, 0.04 %).

Appendix B. Method comparisons

These comparisons justify each production choice; none of them changes a conclusion.

B.1 Band-window determination

Production method (since 2026-08-05): locate bands on the **photon-event column histogram** (the out-of-band floor is flat Poisson; the analog profile' s fixed-pattern baseline is not thresholdable), folded into 10-px sub-windows; baseline and σ from the lowest-40 % sub-windows; bands = runs above baseline+5σ with baseline+2σ gap bridging; runs touching the detection-column edge are discarded as truncation artifacts.

- **vs the 26-07-31 four-Gaussian rule.** A constant-background + four-Gaussian fit of the main-batch histogram converges to μ = 415.9/623.7/845.0/1061.2 px (≤1.5 px from the window centers), σ ≈ 34.5–37.9 px. The production windows are ≈ μ±(1.0–1.45)σ —peak-core *integration* windows; the 26-07-31 μ±2σ rule is a peak *attribution* criterion (deliberately wide). Ported verbatim, that rule would give 170–190 px windows (2–3× wider) and push the residual-noise fraction above 50 %.
- **S/N optimality.** Width scans of $S/\sqrt{(S+N)}$ place the production windows within 0.6 % of optimum (a later all-rows refinement: ∼1 % for B2/B4, ∼6–7 % for the narrower B1/B3 —the basis of the fdr-snr recommendation, not a retroactive change).
- **The 8-method study** (`DOCS/26-08-08-band-window-methods.md`): eight candidate detectors × both batches, plus parameter-plateau, growth-curve, and synthetic-shelf benchmarks, recommend a local-floor Poisson-FDR detection + S/N-optimal width ("fdr-snr") as the future default;

it reproduces the manually-set supplement windows without hand-tuning (FOM within 0.1 %). The study also dissected the production method's supplement failure: B1's over-threshold run touches detection column 350 and the truncation filter discards the whole band (a 2026-08-10 rerun with the edge moved to 330 confirmed the pile-up follows the boundary —the method, not the edge position). Both batches' finalized windows predate the study and stand unchanged.

- **Conclusion stability:** every downstream result was re-checked under the alternative window semantics where applicable (§6.3 dataset A uses S/N-optimal windows) —conclusions unchanged.

B.2 Time-zero estimator

Centroid vs CFD50 on all 19 992 paired frames: centroid − CFD50 = +2.78 px (+2.73 ps) mean —the centroid is dragged late by the pulse's heavy tail —with 1.37 px (1.34 ps) per-frame scatter from shape breathing. The injection experiment (§4) shows the same numbers against ground truth (CFD50 −0.10 ± 0.33 px; centroid +2.71 ± 1.25 px) and that the recovered oscillation period is unaffected. The null result was explicitly re-verified under the centroid variant (UL95 0.013–0.020 vs 0.012–0.019). CFD50 is the production choice on principle (constant-fraction timing is the correct answer to amplitude/shape walk) plus this evidence.

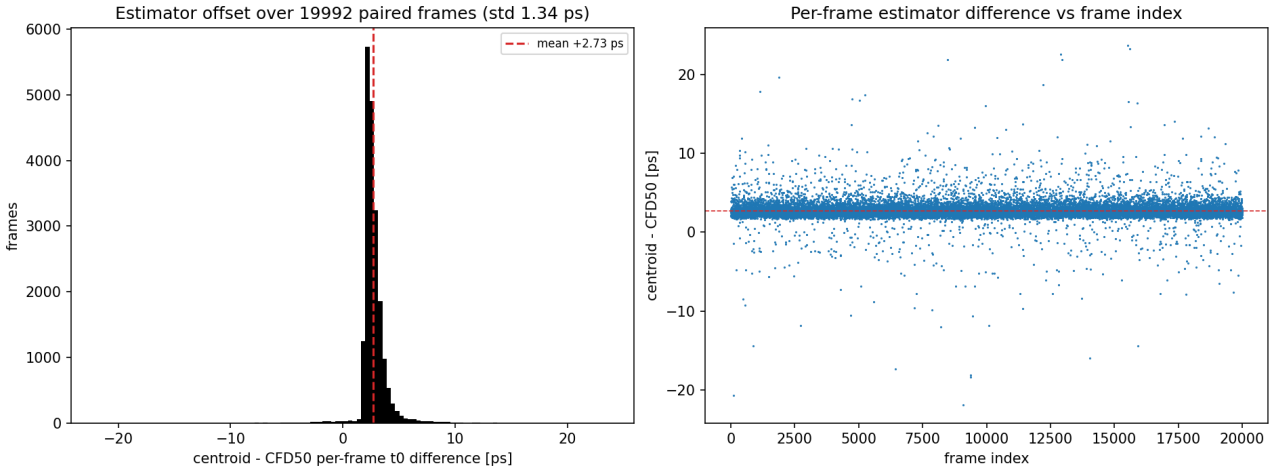


Figure 12: t0 method comparison

Centroid vs CFD50 on all 19 992 paired frames: the per-frame estimator difference (mean +2.73 ps, std 1.34 ps) is stable across the full acquisition.

B.3 Event localization

Production: trackpy (Crocker–Grier) count-weighted centroid, diameter 7, minmass 300. Against circular 2-D Gaussian fits on 1 518 peaks (26-07-31): 92.2 % agree within 0.2 px, median 0.052 px, no systematic offset; Gaussians fail on non-Gaussian elongated MCP clouds (edge collisions), centroids do not. Error budget in §5.

B.4 Residual-noise model and dataset A

Old pipeline: β from the **global** mean out-of-band density. Investigation showed the noise floor is a stable fixed pattern in x (block correlation 0.97), so a global mean misestimates per-band floors by ±30–50 %. Dataset A uses **per-band local-linear floors** (35 px fit windows, 30 px guard bands against the

measured 25–30 px signal tails; B1 one-sided toward the detection edge), ≈20 % lower on average; floor variants bracket the systematic. Effect on the headline numbers: pooled $g^{(2)}(0)$ L 0.531 (local) vs 0.307 (global) —the bracket is smaller than the ±0.38 statistical error —and the old-global variant reproduces the pre-curation numbers nearly bitwise, confirming the estimator chain. Curation itself: out-of-band, defective-column (main: col 676; supplement: 11 hot columns), and frame-edge-touching events removed; morphology cuts were evaluated and rejected (signal and noise are the same MCP-cloud objects; best discriminator 1.7:1 vs 3:1 break-even).

B.5 What does not help

Raising the detection threshold (25→70 counts): −11 % noise at −6 % signal, figure of merit unchanged. Narrowing the band windows around the cores: 1–4 % gain. Both probed on 500-frame / sub-window studies; the only effective lever is acquisition-side background reduction (§5).

Appendix C. Reproduction commands

Each stage' s full command set lives in its execution document (Appendix D): conversion (`DOCS/data.md`, derived-directory READMEs), fig2a pipeline (`DOCS/26-08-04-fig2a.md` § 复现), fig2c pipeline incl. power gate / science tests / injection (`DOCS/26-08-04-fig2c.md` § 复现), supplement batch (`DOCS/26-08-06-hbt-supplement.md` § 复现), dataset A (`DOCS/26-08-08-dataset-a.md`). The report figures are regenerated from the worktree root with:

```
.\.venv\Scripts\python.exe analysis\26-08-10-project-report\render_report_data_preview.py
.\.venv\Scripts\python.exe analysis\26-08-10-project-report\render_report_injection.py
.\.venv\Scripts\python.exe analysis\26-08-10-project-report\render_report_summary.py
.\.venv\Scripts\python.exe `
  analysis\26-08-04-derived-hbt-time-reference\measure_kspace_fringe_oscillation.py `
  --h5 <derived H5> --out analysis\26-08-10-project-report\figures\fig_kspace_fringe.png
```

The PDF version (`DOCS/26-08-10-project-report.pdf`) is built from the worktree root with pandoc + XeLaTeX; all engine/font/filter settings live in `analysis/26-08-10-project-report/pdf_build.yaml` (Cambria covers the Greek/symbol inventory, Microsoft YaHei the two CJK anchor characters above, and a Lua filter lets long code spans wrap inside table cells):

```
pandoc docs\26-08-10-project-report.md -o docs\26-08-10-project-report.pdf `
  --defaults=analysis\26-08-10-project-report\pdf_build.yaml
```

Appendix D. Sub-report index

document	role in this report
`DOCS/adiyatullin2017-fig2a.md` / `adiyatullin2017-fig2c.md`	paper-side fact archive + target decomposition (G-criteria) —§1 sources
`DOCS/26-08-04-fig2a.md`	fig2a pipeline, band windows, null + sensitivity —§3, §6.1
`DOCS/26-08-04-fig2c.md`	power gate, pre-registered tests, upper-limit report —§3 step 8, §6.2, App. A
`DOCS/26-08-06-hbt-supplement.md`	72 000-frame supplement batch —§6.3
`DOCS/26-08-08-dataset-a.md`	dataset A curation + downstream re-analysis —§6.3, App. B.4

document	role in this report
DOCS/26-08-08-dataset-a-noise-model.md	noise-floor structure —§5, App. B.4
DOCS/26-08-08-band-window-methods.md	8-method band-window study —App. B.1
DOCS/08-04_sync_oscillation.md	time-zero estimator argument —App. B.2
DOCS/sem-limitations.md	SEM factorization + frame-count table —§5
DOCS/peak-localization.md / peak-localization-error.md	localization method + error budget —§5, App. B.3
DOCS/measurement-claims.md	claim/evidence boundaries —§1.1, §8.2
DOCS/streak-time-axis.md + metadata-conventions.toml	time-axis conventions —§2.1
DOCS/data.md	data integrity and environment conventions —§2.1
DOCS/experiment-plan.md / literature.md	project gates; literature status (no independent reproduction of the 2017 result found)

Appendix E. Figure provenance

figure	source	generator
Fig. 3 (C1) data preview	this report	analysis/26-08-10-project-report/render_report_data_preview.py (inputs: derived H5 + curated event table)
Fig. 4 (C2) injection	this report	render_report_injection.py (re-runs the frame-level injection; $g^{(2)}$ panel reads drafts/26-08-06_fig2c/fig2c_injection_report.json)
Fig. 5 (C3) sensitivity	drafts/26-08-06_fig2c/power_gate_ul95_vs_period.png (copied)	fig2c_power_gate.py
Fig. 6 (C4) fig2a replica	analysis/26-08-04-derived-hbt-time-reference/fig2a_replica.png	render_fig2a_comparison.py
Fig. 8 (C5) fig2c replica	analysis/26-08-04-derived-hbt-time-reference/fig2c_replica.png	render_fig2c_comparison.py
Fig. 9 (C6) k-space fringes	re-rendered into this report	measure_kspace_fringe_oscillation.py
Fig. 10 (C7) dashboard	this report	render_report_summary.py (numbers with per-entry provenance comments)
§1.2 paper crops	fig2a_paper_original.png / fig2c_paper_original.png	tmp/paper_crop/ (300 dpi from the in-repo PDF)
§6.1 offset-zoom	fig2a_offset_zoom.png	render_offset_zoom.py

figure	source	generator
§6.3 supplement panels	`fig2a_replica_supplement.png`, `fig2c_replica_supplement_bin4px.png`	same renderers, supplement npz/cubes
§6.1 z upper-limit spectrum	`figures/fig_z_upper_limit.png` (copied from `drafts/26-08-06_fig2a_cfd50/`)	`tmp/fig2a_upper_limit_cfd50.py`
App. A.2 test panel	`figures/fig2c_science_tests_bin4px.png` (copied from `drafts/26-08-06_fig2c/`)	`fig2c_science_tests.py`
App. B.2 t0 comparison	`figures/fig_t0_method_comparison.png` (copied from `drafts/26-08-06_fig2a_cfd50/`)	`tmp/fig2a_t0_method_comparison.py`

Appendix F. Glossary

- **HBT (Hanbury Brown–Twiss)**: 50:50 beamsplitter + two detector arms; cross-arm coincidences give the normally ordered intensity correlation.
- **$g^{(2)}(0)$**: zero-delay second-order correlation; 1 = Poisson, >1 = bunched (super-Poisson), <1 = antibunched.
- **CFD / CFD50**: constant-fraction discrimination —timing from the 50 %-of-peak crossing of a pulse's leading edge, immune to amplitude walk and shape breathing.
- **MCP cloud**: the multi-pixel charge cloud one detected photon leaves on the streak tube's microchannel plate intensifier; the event detector integrates it back into one event.
- **Streak camera / synchroscan**: the detector converts arrival time to vertical position (0.984 ps/px here); synchroscan mode accumulates many pulses per exposure in sync with the laser, single-sweep mode records one pulse per frame.
- **t0**: per-frame time zero from the reference pulse; $\tau = y - t0$ is the aligned time coordinate.
- **SEM / jackknife**: standard error of the mean; leave-one-frame-out jackknife is the resampling estimate used for pooled values.
- **Plug-in SEM**: analytic null-hypothesis error band used for per-bin display —the experiment's resolution, not a pointwise error bar.
- **UL95**: one-sided 95 % upper limit, here $|A| + 2\sigma$ on a fitted amplitude.
- **Surrogate**: null-hypothesis replica data (here: circular frame shifts) passed through the identical statistic pipeline.
- **Pre-registration / power gate**: decision criteria written before the computation; a rate-based resolvability estimate computed before the estimator ran.
- **β correction**: subtraction of the residual-background coincidence contribution, estimated from out-of-band events.
- **Dataset A**: the curated event subset (in-band, non-defective, non-edge events) with per-band local noise floors —the "everything kept can be signal" curation.
- **FDR**: false-discovery-rate (Benjamini–Hochberg) thresholding; part of the recommended future band detector.
- **CSI**: Cauchy–Schwarz inequality of intensity correlations —a classical-field test, out of scope here.
- **condensate / mesa**: the two coupled traps holding the condensates are etched circular mesas in

the paper's sample; in our platform the corresponding objects are two reservoir-fed condensates (left/right, or 1/2). This report says "condensate" for our own data and keeps "mesa" only where the paper's sample is described. **reservoir**: the incoherent exciton bath feeding them under nonresonant pumping; **Stark kick**: the optical-Stark pulse that triggers the dynamics in our platform.